

Sitota Ezra Mersha

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SUMMARY

I am a PhD student and researcher passionate about combining my software engineering and research expertise to solve complex challenges in AI acceleration and hardware-software co-design. My research focuses on memory acceleration in disaggregated server environments, chiplet-based architectures, and AI hardware solutions for resource-constrained systems like autonomous vehicles. I excel in collaborative environments, problem-solving, and leadership.

EDUCATION

University of Michigan

PhD in Computer Science and Engineering (2024 - 2029)

- Advisor: Prof. Valeria Bertacco
- Relevant Coursework: Microarchitecture, Advanced Compilers, Advanced Scalable Systems, Artificial Intelligence Foundations, Privacy-enhanced Technologies (PETs)
- GPA: 4.0

Addis Ababa Institute of Technology

Bachelor's Degree in Software Engineering (2018 - 2023)

- Computer Architecture and Organization, Operating Systems, Data Structures and Algorithms
- GPA: 3.84

EXPERIENCE

Graduate Student Research Assistant

University of Michigan

August 2024 – Present

- Achieved 30× faster memory access via data-oblivious algorithms for distributed systems.
- Reduced CPU inference response time by 20% through batching and request aggregation.
- Improved Verilog RTL correctness by 50% for RISC-V pipeline generation.

Full-Stack Drupal Developer and Consultant

International Maize and Wheat Improvement Center (CIMMYT)

April 2023 - April 2024

- Designed a scalable Learning Management System, improving training efficiency by 30%.
- Optimized database queries, reducing system latency by 25%.
- Collaborated with global teams to ensure seamless integration in an agile environment.

Software Engineering Intern

International Livestock Research Institute (ILRI)

Mar 2022 – Oct 2023

- Built a mobile app and web dashboard, reducing manual workflows by 80%.
- Improved API efficiency, reducing response times by 35% for 500+ users.

Software Engineer

Gebeya Solutions

Apr 2022 – Jul 2024

- Developed an e-commerce platform enabling centralized product searches.
- Optimized app memory usage by 20%, improving load times by 15%.

RESEARCH PROJECTS

From Prompt to Processor: LLMs in RTL Design

- Improve Verilog RTL correctness by 44–50% by evaluating 3 LLMs on RISC-V pipeline generation using structured vs. descriptive prompts.
- [Poster](#)

Batching and Request Aggregation for LLM Inference in CPUs

- Optimized request aggregation and batching in Llama.cpp, achieving a 20% speedup in response time for CPU-based inference.
- [Project Report](#) | [Poster](#)

Characterizing Oblivious RAMs

- Implemented and evaluated the performance of several Oblivious RAM (ORAM) techniques to mitigate memory access pattern-based side-channel attacks.
- [Project Report](#)

SYNGEN: A Synthetic Machine Learning Dataset Generator and Expander

- Conducted a study on dataset expansion and generation using machine learning techniques.
- [Presentation Video](#) | [Poster](#)

Rescuing the Future of Memory with Data-Oblivious Algorithms

- Developed a C++ and Python-based simulator to analyze memory access patterns and enhance disaggregated server architecture performance using data-oblivious algorithms.
- [Presentation Video](#) | [Project Report](#) | [GitHub Repository](#)

PROJECTS

Ethiopian Exam Generation and Testing System

- A web application to generate and administer national examinations (for use by about 650,000 students), developed as a capstone project for a Bachelor's degree.
- [GitHub Repository](#)

Lisan

- A flutter application as a complement to therapy for children with speech and language delays, with a focus on an Amharic version to guarantee cultural compatibility.
- [App Demo](#)

Gebeya Dash

- Developed a React Native app for inventory and sales management to help small and medium business owners in Ethiopia improve financial tracking.
- [Project Website](#) | [App Demo](#)

TECHNICAL SKILLS

Languages

Python, C++, C#, Java, Javascript, PHP

Web & Mobile

ReactJS, Flutter, Node.js, Drupal, VueJS, React Native, Laravel

Database & Cloud Platforms

MySQL, PostgreSQL, MongoDB, Redis, AWS, GCP, Heroku, Netlify

Systems & Tools

Scalable storage, performance optimization, Git, CI/CD pipelines